MODULE 25 · DOMAIN 5 - GOVERNANCE, SAFETY AND RISK MANAGEMENT

# Bias, Fairness, Transparency and the 4D Lens

Where bias enters, metrics that look fair but are not, testing outcome disparity, transparency and accountability, and Anthropic's 4D framework.

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admin@sthithapragna.com

Two driving examiners with the same pass rate. **One of them fails the wrong people.**

# Where bias enters a Claude system

- **Pretraining:** the model learned from text that reflects society's patterns.
- **Prompts and examples:** skewed examples teach skewed behaviour.
- **Retrieval and data:** corpora and eval sets that miss some groups.
- **Proxies:** postcode, name or writing style standing in for protected traits.
- **Feedback loops:** today's decisions become tomorrow's training and eval data.

> ON THE EXAM
>
> Bias is a system property. Look at every stage, not just the model.

# Proxies: deleting the field is not enough

| FEATURE | CAN STAND IN FOR | WHAT TO DO |
|---|---|---|
| Postcode | Ethnicity, income | Test outcomes by area |
| Name | Ethnicity, sex | Redact before Claude sees it |
| Writing style or dialect | Nationality, disability, education | Test non-standard English |
| Contact time and channel | Caring duties, working pattern | Justify it, or drop it |

THE TIEBREAKER

Fairness through unawareness fails. Proxies carry the signal back in.

# What Anthropic's own study found

- **Method:** 70 decision scenarios, varying only demographic details.
- **Finding:** patterns of both positive and negative discrimination, without interventions.
- **Mitigation:** careful prompt engineering significantly reduced both.
- **Position:** Anthropic does not endorse or permit automated decisions in these high-risk uses.

ON THE EXAM

Test your own system the same way: change only the demographic detail, and compare the outcome.

# Metrics that look fair, and what they miss

| METRIC | WHAT IT SHOWS | WHAT IT MISSES |
|---|---|---|
| Overall accuracy | How often the system is right | Who bears the errors |
| Approval-rate parity | Equal approval rates across groups | Different error rates behind them |
| Agreement with past decisions | Matches what people did before | Reproduces their bias |
| Error-rate parity | False approvals and declines per group | Needs labelled outcomes per group |

THE TIEBREAKER

**Approval parity is not error parity. Agreement is not fairness.**

# Outcome-disparity testing, in order

01
### Define
Groups, outcomes, harms that matter.

02
### Build cases
Representative, plus counterfactual pairs.

03
### Measure
Outcomes and error rates per group.

04
### Investigate
Which stage causes each gap?

05
### Mitigate
Prompt, data, process, human review.

06
### Re-test
On every change, as a regression.

ON THE EXAM

Disparity tests belong in the regression suite. Every change re-runs them.

# Mitigations that work, and ones that only look like it

THESE REDUCE DISPARITY

## Change the system

- Redact names and irrelevant personal details
- Instruct Claude to ignore protected traits
- Rebalance examples and evaluation sets
- Test per group, after every change
- Route borderline cases to trained people

THESE ONLY LOOK LIKE IT

## Change the story

- Deleting the protected field alone
- Reporting overall accuracy
- Asking Claude whether it was fair
- A fairness statement in the policy
- Testing once, before launch

THE TIEBREAKER

A mitigation counts only when a disparity test shows the gap closing.

# Transparency and accountability

- **Disclose AI involvement:** at the start of a session, and to anyone a decision affects.
- **Offer a path to a human,** easy to find and honoured quickly.
- **Explain the basis:** sources, reasons, what the system cannot do.
- **Accountability stays with people:** a named owner, decision rights, an appeal route.
- **Record decisions** so they can be audited and challenged.

> ON THE EXAM
>
> The model is never the accountable party. A named person, with authority, always is.

# Explanations you can stand behind

- **Stated reasoning is not proof:** Claude 3.7 Sonnet mentioned a planted hint only 25% of the time.
- **Record the facts:** inputs, sources retrieved, prompt and model versions, the output.
- **Reasons come from the decider:** the person who approves states why.
- **Explain in plain language** what the decision rested on, and how to challenge it.

> ON THE EXAM
>
> Use stated reasoning to debug. Use records and human reasons to explain.

# The 4D lens, applied by an architect

| COMPETENCY | ANTHROPIC'S DEFINITION | THE ARCHITECT'S ACTION |
|---|---|---|
| Delegation | Deciding what work to do with AI | Split tasks by type of risk |
| Description | Communicating clearly with AI | Prompts, criteria, context, examples |
| Discernment | Evaluating outputs and behaviour critically | Evals, review, monitoring |
| Diligence | Interacting with AI responsibly | Disclosure, accountability, safe deployment |

ON THE EXAM

Three modes of working sit alongside: automation, augmentation and agency. Modes are not competencies.

# An ethics checklist before launch

- **Purpose:** who benefits, who is affected, who could be harmed?
- **Disparity:** tested across the groups served, with results recorded?
- **Transparency:** AI disclosed, and a human reachable?
- **Accountability:** a named owner, and an appeal route?
- **Monitoring:** complaints and disparities watched after launch?

ON THE EXAM

A checklist turns good intentions into a launch gate. Unanswered questions block release.

> Ashgrove Insurance's claims assistant recommends fast-track payment or manual review. Fast-track rates are equal across age groups.
>
> But claims handlers report that legitimate claims from older customers are often sent to slow manual review.

## Which analysis should Corvane add?

A. None, because fast-track rates are already equal across age groups

B. Error rates per age group: legitimate claims sent to review, and invalid claims fast-tracked

C. Overall accuracy across all claims

D. Agreement between the assistant and past handler decisions

◆ PAUSE AND DECIDE

## Which analysis should Corvane add?

A None, because fast-track rates are already equal across age groups

*Equal rates can hide unequal errors.*

B Error rates per age group: legitimate claims sent to review, and invalid claims fast-tracked

Shows who bears the mistakes.

C Overall accuracy across all claims

*Hides how errors fall across groups.*

D Agreement between the assistant and past handler decisions

*Would reproduce any past bias.*

THE SIGNAL

**Approval parity is not error parity. Measure who bears the errors.**

> Mereside's complaints triage agrees with historical handler classifications 94% of the time. The team calls it fair.
>
> An audit shows handlers used to downgrade complaints written in non-standard English.

## What should Corvane recommend?

A. Accept the agreement rate as evidence of fairness

B. Raise the agreement target to near total

C. Test outcomes by writing style against an expert-relabelled set, with counterfactual rewrites of the same complaints

D. Stop the assistant from reading the complaint text at all

◆ PAUSE AND DECIDE

## What should Corvane recommend?

A Accept the agreement rate as evidence of fairness

*Agreement with biased decisions copies the bias.*

B Raise the agreement target to near total

*Optimises towards the bias.*

C Test outcomes by writing style against an expert-relabelled set, with counterfactual rewrites of the same complaints

Fresh labels and controlled rewrites isolate the effect.

D Stop the assistant from reading the complaint text at all

*It cannot triage without the text.*

THE SIGNAL

Agreement with past humans is not fairness. Relabel independently and test with counterfactuals.

> Quillon's clinicians worry that prior-authorisation letters are less persuasive for some patients than others.
>
> Corvane must design a test for whether patient demographics change the letters.

## Which test design is best?

A. Ask Claude whether its letters are biased

B. Counterfactual pairs: identical cases varying only name, age or language, with letters compared by calibrated graders

C. A random sample of letters reviewed by one clinician

D. Remove all demographic fields from the inputs and assume the problem is solved

◆ PAUSE AND DECIDE

## Which test design is best?

A Ask Claude whether its letters are biased

*Self-report is not measurement.*

B Counterfactual pairs: identical cases varying only name, age or language, with letters compared by calibrated graders

Isolates the demographic effect, at scale.

C A random sample of letters reviewed by one clinician

*Different cases, one opinion. The effect is confounded.*

D Remove all demographic fields from the inputs and assume the problem is solved

*Proxies remain, and age can be clinically relevant.*

THE SIGNAL

To test a demographic effect, change only the demographic detail. Everything else stays identical.

> Tallyhawk's support agent introduces itself as Sam from support and never says it is AI.
>
> Customers who ask for a person are told the agent can handle everything.

## What should Corvane recommend?

A Keep it, since the persona improves satisfaction scores

B Mention AI use in the terms of service

C Disclose AI at the start of each session, and offer a clear, honoured path to a person

D Keep the human name, but escalate when customers insist repeatedly

◆ PAUSE AND DECIDE

## What should Corvane recommend?

A Keep it, since the persona improves satisfaction scores

*Breaks the Usage Policy's disclosure requirement.*

B Mention AI use in the terms of service

*Not disclosure at the start of each session.*

C Disclose AI at the start of each session, and offer a clear, honoured path to a person

Meets the policy, and respects the customer.

D Keep the human name, but escalate when customers insist repeatedly

*Still implies a person; the path should not need insisting.*

THE SIGNAL

Customers must know they are talking to AI, and be able to reach a person without a fight.

> Mereside's chief operating officer asks who is accountable if an AI-drafted complaint response sent to a customer turns out to be wrong.
>
> Handlers approve every draft before it is sent.

## What should Corvane's design state?

A Mereside, through the approving handler, under a named owner with decision rights and an appeal route

B Anthropic, as the provider of the model

C Corvane, as the firm that built the system

D Nobody, because AI output is only advisory

◆ PAUSE AND DECIDE

## What should Corvane's design state?

A. Mereside, through the approving handler, under a named owner with decision rights and an appeal route
Accountability stays with the people who decide.

B. Anthropic, as the provider of the model
*Providers supply tools; deployers own decisions.*

C. Corvane, as the firm that built the system
*Builders hand over; the business owns operation.*

D. Nobody, because AI output is only advisory
*An accountability gap the regulator will not accept.*

THE SIGNAL

Diligence in practice: accountability stays with named people who have the authority to decide.

> Quillon's appeals team wants an official reason stored for each coverage recommendation that a clinician signs off.
>
> They propose saving Claude's stated reasoning as that reason.

## What should Corvane recommend?

A. Save Claude's stated reasoning, since it shows how the model decided

B. Turn reasoning off, so there is nothing to misread

C. Ask Claude to explain its first answer again, and store that

D. Record inputs, sources and versions, with the clinician's own reasons as the official record

◆ PAUSE AND DECIDE

## What should Corvane recommend?

A Save Claude's stated reasoning, since it shows how the model decided

*Stated reasoning is not always faithful.*

B Turn reasoning off, so there is nothing to misread

*Loses a debugging aid, and still no reason.*

C Ask Claude to explain its first answer again, and store that

*A second story is still not the decision basis.*

D Record inputs, sources and versions, with the clinician's own reasons as the official record

Facts on record, reasons from the decider.

THE SIGNAL

The person who signs the decision gives the reason. The record proves what the system saw.

> Corvane's review of a client's design finds two gaps. Nobody evaluated the agent's outputs critically before launch.
>
> And the team never decided which tasks the agent should not do at all.

**Which two 4D competencies do these gaps fall under? Choose two.**

A It falls under Discernment

B It falls under Description

C It falls under Automation

D It falls under Delegation

E It falls under Agency

◆ PAUSE AND DECIDE

## Which two 4D competencies do these gaps fall under? Choose two.

A It falls under Discernment
Evaluating outputs critically.

B It falls under Description
*Communicating clearly with AI. Not the gap here.*

C It falls under Automation
*A mode of working, not a competency.*

D It falls under Delegation
Deciding what work AI should and should not do.

E It falls under Agency
*A mode of working, not a competency.*

THE SIGNAL

Four competencies, three modes. Do not mix them up.

# In five lines

- **Bias is a system property:** pretraining, examples, data, proxies, feedback loops.
- **Approval parity is not error parity;** agreement with humans is not fairness.
- **Test disparity:** per-group error rates, counterfactual pairs, re-tested on every change.
- **Disclose AI, offer a human,** and keep accountability with named people.
- **4D:** Delegation, Description, Discernment, Diligence. Modes are not competencies.

COMING UP

Domain 5 complete. Next, Domain 6: structured discovery and requirements.